

Table of Contents:

Topic	Page Number
Introduction	3-4
Project Graph	5
List of Objects	5-6
List of Functions (Represent Object)	6-7
List of Animation Function	7
Conclusion	8

1. Introduction

Scene-1: Urban Commute – Dhaka City

This project successfully demonstrates the design and implementation of an interactive urban city scene inspired by Dhaka City using OpenGL and GLUT. The scene integrates multiple real-world elements such as bridges, roads, educational institutions, hospitals, residential buildings, vehicles, rivers, and natural components to create a visually structured environment.

A wide range of animations have been implemented, including moving cars, buses, trains, ships, clouds, water waves, rainfall, snowfall, and dynamic sun–moon transitions for day–night and sunset effects. These animations enhance realism and show effective use of timer-based functions for continuous motion.

The project also incorporates mouse interactions, allowing users to toggle weather conditions such as rain and snow, and keyboard interactions to control scene behaviour including day/night switching, sunset animation, cloud movement, and train operation. This interactivity makes the scene engaging and demonstrates practical event-driven programming.

Overall, the project fulfils its objectives by combining object-based scene design, smooth animations, and user interactions within a single integrated environment. It reflects a clear understanding of OpenGL primitives, transformations, animation logic, and interaction handling, making it a strong example of an interactive computer graphics application.

Objects

The scene contains multiple graphical objects, including but not limited to:

- Bridge and bridge pillars
- Road with lane markings
- School building with doors and windows
- Hospital and hospital section
- Residential and commercial buildings (total 5)
- Grass total 3
- Tree total 4
- Electric wires
- Traffic light
- Vehicles such as cars (red orange, blue), bus, train, and ship
- waves

- Natural elements like river, cloud, sun, moon, snow, rain

Animations

Several animation functions are implemented to simulate real-world motion:

- Moving vehicles (cars, bus, train, ship)
- Cloud movement across the sky (left and right)
- Wave motion in the river
- Day–night transition with sun and moon animation and Sunset animation
- Weather effects including rain and snowfall

Mouse Interaction

Mouse interaction is used to control weather effects:

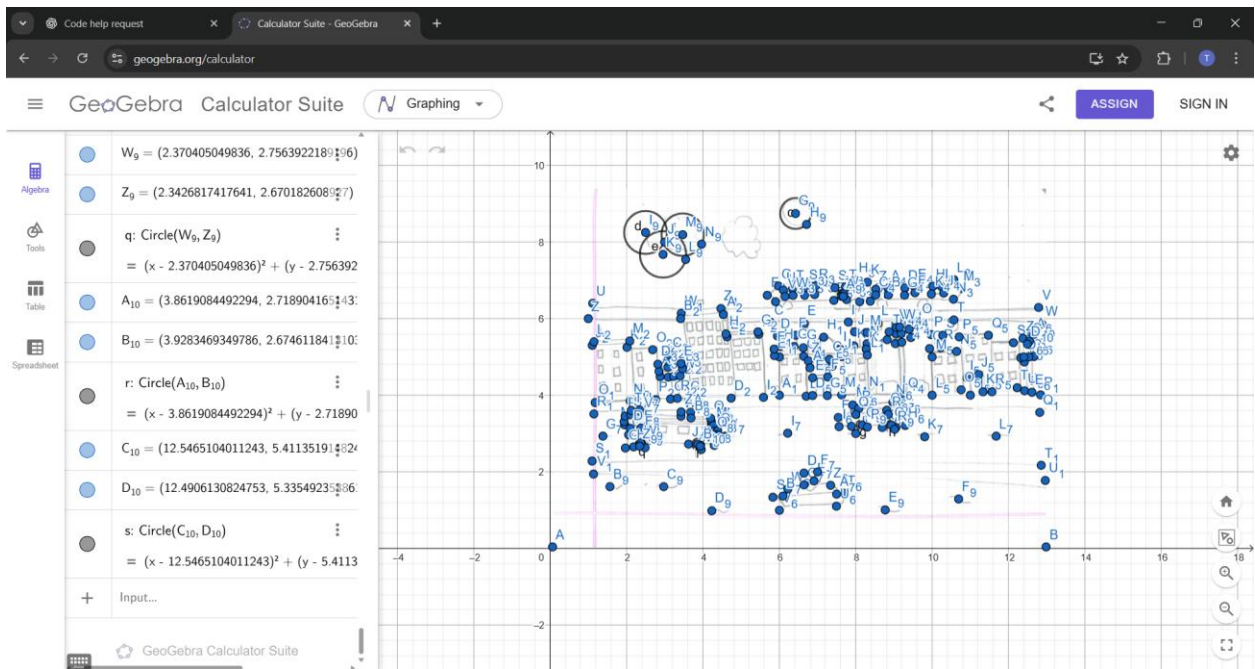
- **Left mouse click** toggles the rain animation
- **Right mouse click** toggles the snow animation

Keyboard Interaction

Keyboard input is used to control scene behaviour and animations:

- **N / n** → Toggle between day and night mode
- **R / r** → Start sunset animation
- **M / m** → Start train movement
- **S / s** → Stop train movement
- **L / l** → Move clouds left to right
- **J / j** → Move clouds right to left
- **K / k** → Stop cloud movement

2. Graph:



3. List of Object:

SL#	Object ID	Object Name
1	0	Bridge
2	1	Bridge Pillar 1
3	2	Bridge Pillar 2
4	3	Bridge Pillar 3
5	4	School Building
6	5	School Door
7	6	School Windows
8	7	Hospital
9	8	Hospital Section
10	9	Residential Building 1
11	10	Road
12	11	Grass1
13	12	Grass2
14	13	Grass3
15	14	Electric Wire
16	15	Car Red

17	16	Car 2
18	17	Car 3
19	18	BRTC Bus
20	19	River
21	20	Ship
22	21	Train
23	22	Cloud
24	23	Tree
25	24	Residential Building 2
26	25	Building 2
27	26	Building 3
28	27	Building 4
29	28	Traffic Signal
30	29	Sun
31	29	Moon
32	30	Rain
33	31	Snow

4. List of Functions To Represent Objects- Each object must be represented by a function. MUST follow the following listing pattern-

SL#	Function Name	Object ID
1	sabridge()	0
2	sabridgeP1() + sabridgeP11()	1
3	sabridgeP2() + sabridgeP22()	2
4	sabridgeP3() + sabridgeP33()	3
5	sschool() + sschoolLine()	4
6	sschoolDoor()	5
7	schoolWindowsLeft() + schoolWindowsRight() + bigSchoolWindow()	6
8	shospital()	7
9	shospital_section()	8
10	sbuild() + sbuildline()	9
11	sroad() + dashlineroad()	10
12	sgrass1()	11
13	sgrass2()	12
14	sgrass3()	13
15	swire()	14
16	sdrawCarRed()+drawCircle()	15

17	sbluecar2()	16
18	scar3()	17
19	sdrawBRTCbus()	18
20	sriver()	19
21	sship()	20
22	strain()	21
23	scircle()+sdrawCloud+ drawAllClouds()	22
24	drawTree()+drawTree()+drawTree()	23
25	building1()	24
26	building2()	25
27	Building3()	26
28	Building4()+buildingBigDoor()	27
29	sdrawTraffic()	28
30	sSun()	29
31	initRain()+void drawRain()+	30
32	initShow()+drawSnow()	31

5. List of Animation Functions with ID – These should be functions that are used to add animations like- object movement, mouse, and keyboard interaction, change of scenarios and so on. MUST follow the following listing pattern-

SL#	Animation Function	Animation Function ID
1	updateAnimation() sun and moon	A1
2	updateBus()	A2
3	updateCarRed()	A3
4	updateCarBlue()	A4
5	updateCarOrange()	A5
6	updateTrain()	A6
7	updateShip()	A7
8	updateCloud()	A8
9	updateRain()	A9
10	updateSnow()	A10
11	updateWave()	A11
12	mouse()	A12
13	Keyboard()	A13

6. Conclusion

This project presents an interactive 2D computer graphics scene developed using OpenGL (GLUT) in C++, titled Urban Commute – Dhaka City.

The scene illustrates a busy city environment featuring roads, bridges, buildings, vehicles, river transport, and weather effects. The composition reflects daily urban life in Dhaka, combining static and dynamic objects to enhance realism.

The system supports **real-time animation** and **user interaction** through keyboard and mouse controls. Features like **day–night transition**, **sunset animation**, **moving vehicles**, **flowing river**, **weather effects (rain and snow)** enhance the realism of the environment.